

From Action Labels to Sets: Rethinking Action Supervision for Imitation Learning from Corrective Feedback

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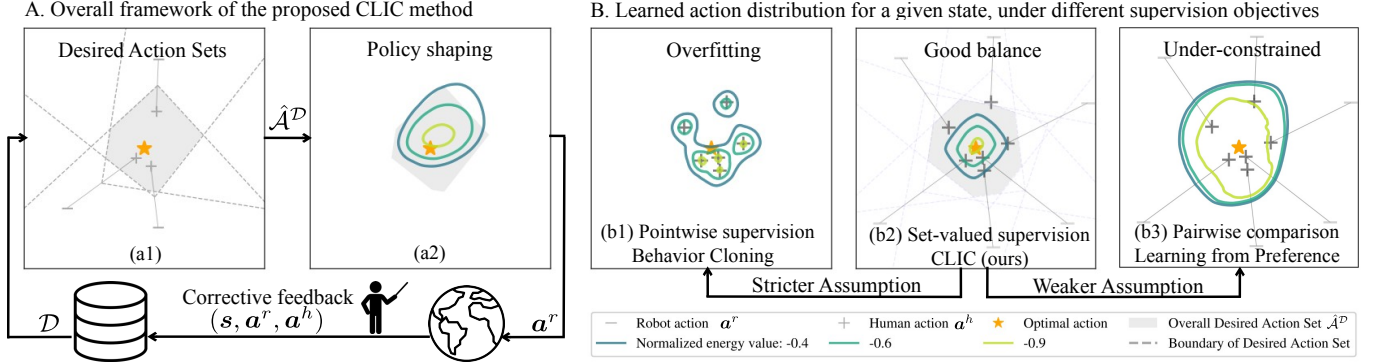


Fig. 1: A: Our method operates in an Interactive Imitation Learning framework. Example rollouts of this framework are shown in Fig. 2. At each step, the robot’s policy outputs an action a^r , and a human teacher provides corrective feedback occasionally if the robot action is suboptimal. In (a1), each feedback stored in the data buffer $\mathcal{D}$ defines a desired action set—actions consistent with the feedback given prior knowledge of its type. These sets are then aggregated to form an overall desired action set $\hat{\mathcal{A}}^{\mathcal{D}}$. In (a2), the policy, represented as an energy-based model (EBM), is trained to generate actions within $\hat{\mathcal{A}}^{\mathcal{D}}$. **B:** Examples of a learned action distribution in a 2D action space. Pointwise BC [3] can overfit individual action labels, while pairwise comparison may provide supervision that is too weak to sufficiently constrain the action space. CLIC uses set-valued supervision, which offers a balance between pointwise targets and pairwise comparisons, recovering the underlying optimal action without overfitting to individual labels or under-constraining the policy.

Abstract—Behavior cloning (BC) optimizes policies by treating human demonstrations as pointwise action labels. While effective with accurate action labels, this formulation is brittle in practice: when human-provided actions are imperfect, treating each label as an exact target can steer the policy away from the underlying desired behavior, particularly when expressive models are used (e.g., energy-based models). As a result, we propose a human-in-the-loop alternative that replaces pointwise supervision with set-valued action targets. We introduce *Contrastive policy Learning from Interactive Corrections (CLIC)*. CLIC leverages human corrections to construct and refine sets of desired actions, and optimizes a policy to place probability mass over these sets rather than over a single action target. This formulation naturally accommodates both absolute and relative corrections and can represent complex multi-modal behaviors. Extensive simulation and real-robot experiments show that the proposed approach leads to effective policy learning across diverse settings: CLIC remains competitive with the state of the art under accurate data while being substantially more robust under noisy, relative, and partial feedback. Our implementation is publicly available at <https://clic-webpage.github.io/>.

I. INTRODUCTION

Behavior Cloning (BC) enables robots to acquire complex skills by imitating human demonstrations [4, 6, 7, 9]. It casts policy learning as supervised learning by treating the demonstrated action at each state as a pointwise training target. This formulation also extends to interactive settings under absolute corrective feedback, where a human intervenes by taking control of the robot in states where the learning policy performs poorly, and uses these corrective actions as training targets to iteratively refine the policy [2, 3, 5]. In practice,

however, human-provided actions—whether collected offline or through interactive corrections—can be suboptimal or noisy. This is problematic because expressive policy classes, derived from, e.g., energy-based models or score functions, are prone to overfitting such suboptimal data rather than recovering the underlying optimal behavior [1, 10].

In contrast, work focusing on learning from preference feedback relaxes the pointwise BC assumption by replacing exact action labels with pairwise comparisons between actions or trajectories [8]. However, such pairwise ranking losses constrain only the relative ordering between two actions, leaving the rest of the action space largely unconstrained. As illustrated in Fig. 1B, pointwise BC can overfit individual action labels, while pairwise comparison may provide supervision that is too weak to sufficiently constrain the action space. This leaves a gap for a supervision objective that is more robust than pointwise BC and more informative than pairwise preference learning.

We propose a set-based formulation that bridges these two extremes. Rather than treating human corrections as pointwise targets or pairwise comparisons, we interpret them as specifying a desired action set. That is, for each correction at a given state, we define a subset of the action space that includes actions consistent with the feedback while excluding inconsistent ones. We then design a training objective that encourages the policy to place probability mass on this desired action set. A single correction is generally insufficient for the policy to recover the optimal action at a given state, since the corresponding desired action set may still contain suboptimal actions. This motivates the interactive nature of our framework, allowing such sets to be iteratively refined as more feedback is provided. Fig. 1A

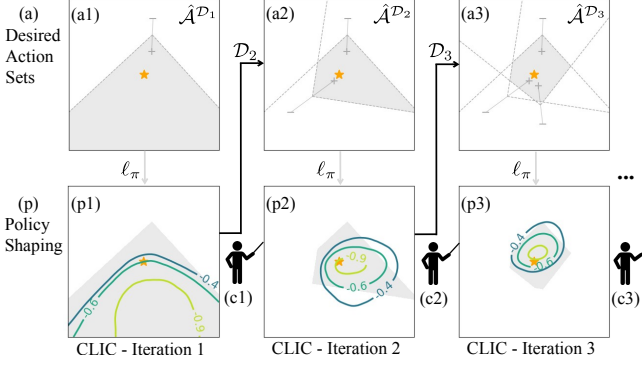


Fig. 2: Overview of CLIC. The gray-shaded area represents desired action sets induced by corrective feedback, while the contour map shows the energy landscape of the EBM policy. As iterations progress, the overall desired action set is gradually refined with new corrective feedback, and the robot policy is updated accordingly.

summarizes the interactive learning framework and the use of desired action sets in policy shaping.

II. METHOD

We propose Contrastive policy Learning from Interactive Corrections (CLIC) to address the limitations of pointwise behavior cloning objectives. CLIC replaces pointwise action targets with set-valued supervision. For each corrective feedback, CLIC first constructs a desired action set. This set, rather than a single action label $\mathbf{a}^h$, represents a region of desired actions that are consistent with this corrective feedback. The aggregation of multiple corrections for a given state leads to a progressively refined overall desired action set, and the policy adapts accordingly.

A. Desired action sets: Formulation and Aggregation

For a corrective feedback $(\mathbf{a}^r, \mathbf{a}^h)$ at state $\mathbf{s}$, we define a desired action set $\hat{\mathcal{A}}(\mathbf{a}^r, \mathbf{a}^h)$ such that $\mathbf{a}^h \in \hat{\mathcal{A}}(\mathbf{a}^r, \mathbf{a}^h)$ and $\mathbf{a}^r \notin \hat{\mathcal{A}}(\mathbf{a}^r, \mathbf{a}^h)$. Let $\mathcal{A}_s^*$ denote the optimal-action set at $\mathbf{s}$. We assume

$$\hat{\mathcal{A}}(\mathbf{a}^r, \mathbf{a}^h) \cap \mathcal{A}_s^* \neq \emptyset, \quad (1)$$

and allow $\hat{\mathcal{A}}(\mathbf{a}^r, \mathbf{a}^h)$ to contain actions outside $\mathcal{A}_s^*$. This formulation makes a weaker assumption than point-label BC: instead of requiring the correction to identify a unique optimal action, it only requires the induced set to remain compatible with at least one optimal action. Meanwhile, it is stronger than the pairwise comparison objective, because it constrains a region of the action space rather than merely preferring one action over another.

The polytope desired action set is constructed by intersecting half-space desired sets. A half-space $\mathcal{A}^H$ can be constructed from a pair of actions $(\mathbf{a}^-, \mathbf{a}^+)$:

$$\mathcal{A}^H(\mathbf{a}^-, \mathbf{a}^+) = \{\mathbf{a} \in \mathcal{A} \mid \mathbb{D}(\mathbf{a}, \mathbf{a}^-) \geq \mathbb{D}(\mathbf{a}, \mathbf{a}^+)\}, \quad (2)$$

where $\mathbb{D}(\mathbf{a}_1, \mathbf{a}_2) = \|\mathbf{a}_1 - \mathbf{a}_2\|$. Given a corrective feedback, multiple contrastive action pairs are generated via data augmentation, and their half-spaces are intersected. The resulting polytope desired action set is applicable to both relative and absolute corrections. When the human feedback is known to be an absolute correction, it provides stronger information and can be used to define a circular desired action set,

$$\hat{\mathcal{A}}^C(\mathbf{a}^r, \mathbf{a}^h) = \{\mathbf{a} \in \mathcal{A} \mid (1 - \varepsilon)\mathbb{D}(\mathbf{a}^r, \mathbf{a}^h) \geq \mathbb{D}(\mathbf{a}, \mathbf{a}^h)\}. \quad (3)$$

For $\varepsilon \rightarrow 1$, the ball reduces to the single action point $\mathbf{a}^h$, recovering pointwise supervision as a limiting case.

During interactive learning, the corrective feedback provided by the teacher is aggregated into a data buffer $\mathcal{D} = \{[\mathbf{s}_t, \mathbf{a}_t^r, \mathbf{a}_t^h], t = 1, \dots\}$. The desired action sets from these feedback signals collectively define an overall desired action set for each state. For the unimodal case, this overall desired action set is the intersection of the individual desired action sets induced by the feedback at that state. This intersection refines the desired region as more feedback is obtained. Rather than explicitly maintaining this set for every state, CLIC realizes the set-valued supervision through a surrogate loss that encourages the learner’s policy to increase the probability of selecting actions within the desired action sets.

B. Policy Shaping via Desired Action Sets

For implicit policies, CLIC updates an energy-based policy $\pi_\theta(\mathbf{a}|\mathbf{s}) \propto \exp(-E_\theta(\mathbf{s}, \mathbf{a}))$ to align with a target distribution that assigns high probability to actions within the desired set. This is done by minimizing

$$\ell_{KL}(\theta) = \mathbb{E}_{(\mathbf{a}^h, \mathbf{a}^r, \mathbf{s}) \sim p_{\mathcal{D}}} [\text{KL}(\pi^{\text{target}}(\mathbf{a}|\mathbf{s}) \parallel \pi_\theta(\mathbf{a}|\mathbf{s}))]. \quad (4)$$

The target distribution is constructed as a posterior distribution using a soft set-membership likelihood and a prior over actions. The observation model $p(\mathbf{a} \in \hat{\mathcal{A}}(\mathbf{a}^r, \mathbf{a}^h) | \mathbf{a}, \mathbf{s})$ quantifies the degree to which an action is consistent with the desired action set. In the noise-free case, this observation model reduces to the indicator function $\mathbb{I}[\mathbf{a} \in \hat{\mathcal{A}}(\mathbf{a}^r, \mathbf{a}^h)]$. To account for noisy human feedback and mismatch in the set construction, CLIC replaces the hard membership boundary with a smooth sigmoid. For a half-space desired action set, this is written as

$$p(\mathbf{a} \in \mathcal{A}^H(\mathbf{a}^-, \mathbf{a}^+) | \mathbf{a}, \mathbf{s}) = \sigma_T(\mathbb{D}(\mathbf{a}, \mathbf{a}^-) - \mathbb{D}(\mathbf{a}, \mathbf{a}^+)), \quad (5)$$

where $\sigma_T(x) = (1 + \exp(-x/T))^{-1}$ and $T > 0$ controls the sharpness of the transition. For a polytope desired set, the observation model is obtained by multiplying the likelihoods of its constituent half-spaces; for a circular desired set, the sigmoid is applied to the distance from the ball boundary. We use the current policy as the prior, yielding

$$\pi^{\text{target}}(\mathbf{a}|\mathbf{s}) \propto p(\mathbf{a} \in \hat{\mathcal{A}}(\mathbf{a}^r, \mathbf{a}^h) | \mathbf{a}, \mathbf{s}) \pi_\theta(\mathbf{a}|\mathbf{s}). \quad (6)$$

For unimodal tasks, CLIC also admits an explicit Gaussian-policy formulation, referred to as CLIC-Explicit.

III. EXPERIMENTS

We evaluate CLIC through simulation and real-robot experiments. In simulation, we compare CLIC with HG-Dagger, D-COACH, IBC, PVP, Diffusion Policy (DP), and Ambient DP (ADP). The tasks include Push-T and three manipulation tasks from robosuite. Each method was run for 160 episodes, and each experiment was repeated three times to calculate final success rates and convergence time steps.

Under accurate absolute feedback, CLIC-Half and CLIC-Circular perform strongly, consistently outperforming IBC and remaining competitive with DP. CLIC-Circular performs best under accurate absolute feedback because its circular desired action sets provide more specific supervision than the polytope sets used by CLIC-Half when the teacher action is accurate. PVP performs poorly on the robosuite tasks, suggesting that

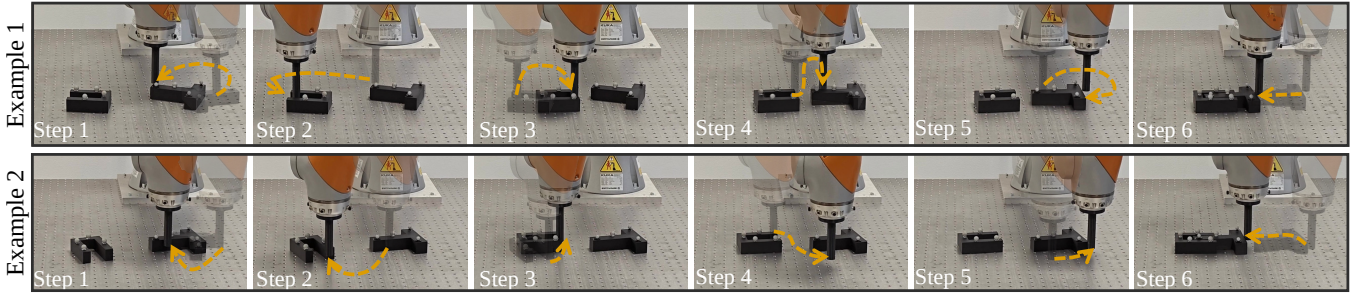


Fig. 3: Examples of CLIC-Half policy rollout for the Insert-T task after training. At each step, the transparent figure shows the initial state, and the orange arrow indicates the end-effector's trajectory. The solid figure shows the end state.

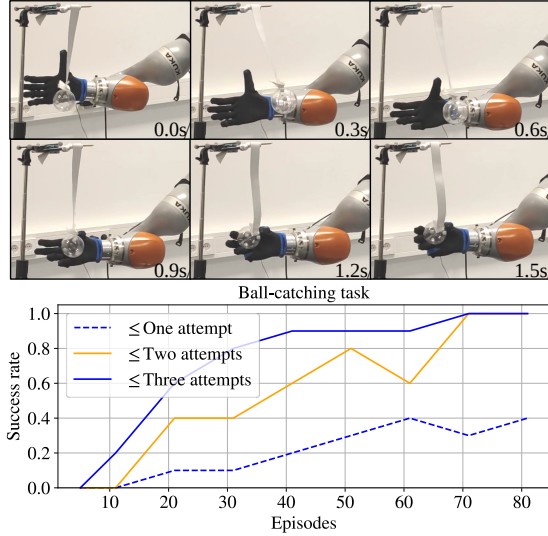


Fig. 4: Experiment results for the ball-catching task.

pairwise comparison alone provides supervision that is too weak to effectively shape an implicit policy.

With noisy demonstrations, CLIC-Half and CLIC-Circular experience much smaller performance drops compared to baselines like DP and IBC. These baselines degrade because of their strict assumption of accurate action labels. ADP outperforms DP but is worse than CLIC-Half and CLIC-Circular in the reported settings. CLIC controls the desired action set through a small number of interpretable hyperparameters, allowing the induced set to better match the uncertainty of the feedback signal.

For partial feedback, CLIC-Half maintains a high success rate on TwoArm-Lift. Human actions consist of feedback dimensions, where human feedback is provided, and non-feedback dimensions, which may be suboptimal. Pointwise methods attempt to imitate the entire teacher action, including suboptimal non-feedback dimensions. In contrast, CLIC-Half imitates a desired action set rather than a single action label. For relative corrections, CLIC-Half and CLIC-Explicit show only small performance drops compared to their results under absolute corrections, while baselines based on the pointwise BC objective fail because the BC loss can mislead policy updates when the current policy's actions are better than human actions in the dataset.

A. Real-robot Validations

In real-robot validations, we evaluate Insert-T, ball catching, and water pouring using a 7-DoF KUKA iiwa manipulator.

Insert-T requires the robot to insert a T-shaped object into a U-shaped object by pushing to adjust their positions and orientations. Compared to the simulated Push-T task, Insert-T is more complex because it involves two objects, introduces multi-modal decisions about which object to manipulate first, and has an increased task horizon. For easy tasks, baseline methods perform similarly to CLIC but converge more slowly. For medium and hard tasks, CLIC achieves significantly higher success rates; in hard tasks, CLIC achieves 80% success compared to 30% for DP and 10% for IBC. Examples of post-training policy rollouts for CLIC are shown in Fig. 3.

The ball-catching task is challenging because of its highly dynamic nature, making successful demonstrations difficult to provide. Instead, relative corrective feedback is more intuitive and easier because humans can occasionally provide direction signals to improve the robot's policy. The robot must coordinate the ball's motion, the end-effector's motion, and the gripper's actuation, making partial feedback suitable. Fig. 4 shows the ball-catching results, including success rates over training and a post-training rollout. By the end of training, the robot achieves a 1.0 success rate for catching the ball within two attempts, and its first-attempt success rate continues to improve to 0.4.

The water-pouring task requires the robot to control the pose of a bottle to precisely pour liquid into a small bowl. The human teacher can switch between absolute and relative corrections via keyboard. Initially, absolute feedback is preferred as the policy is learned from scratch; as the policy improves, relative corrections make it easier to refine the policy in specific regions of the state space. The success rate increases from 0.6 in Episode 26 to 0.9 by Episode 41.

IV. CONCLUSION

In this paper, we introduced set-valued supervision for learning from corrective feedback. Building on this perspective, we proposed CLIC, a practical interactive imitation learning framework that constructs desired action sets from human corrections and trains policies to align with these sets. Across simulation and real-robot experiments, we showed that this formulation is effective across diverse feedback settings: it remains competitive under accurate feedback, is more robust under noisy, inconsistent, relative, and partial feedback, and supports multi-modal behaviors with implicit policies. These results show that set-valued supervision provides a useful middle ground between pointwise behavior cloning and pairwise comparison objectives for interactive policy learning.

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